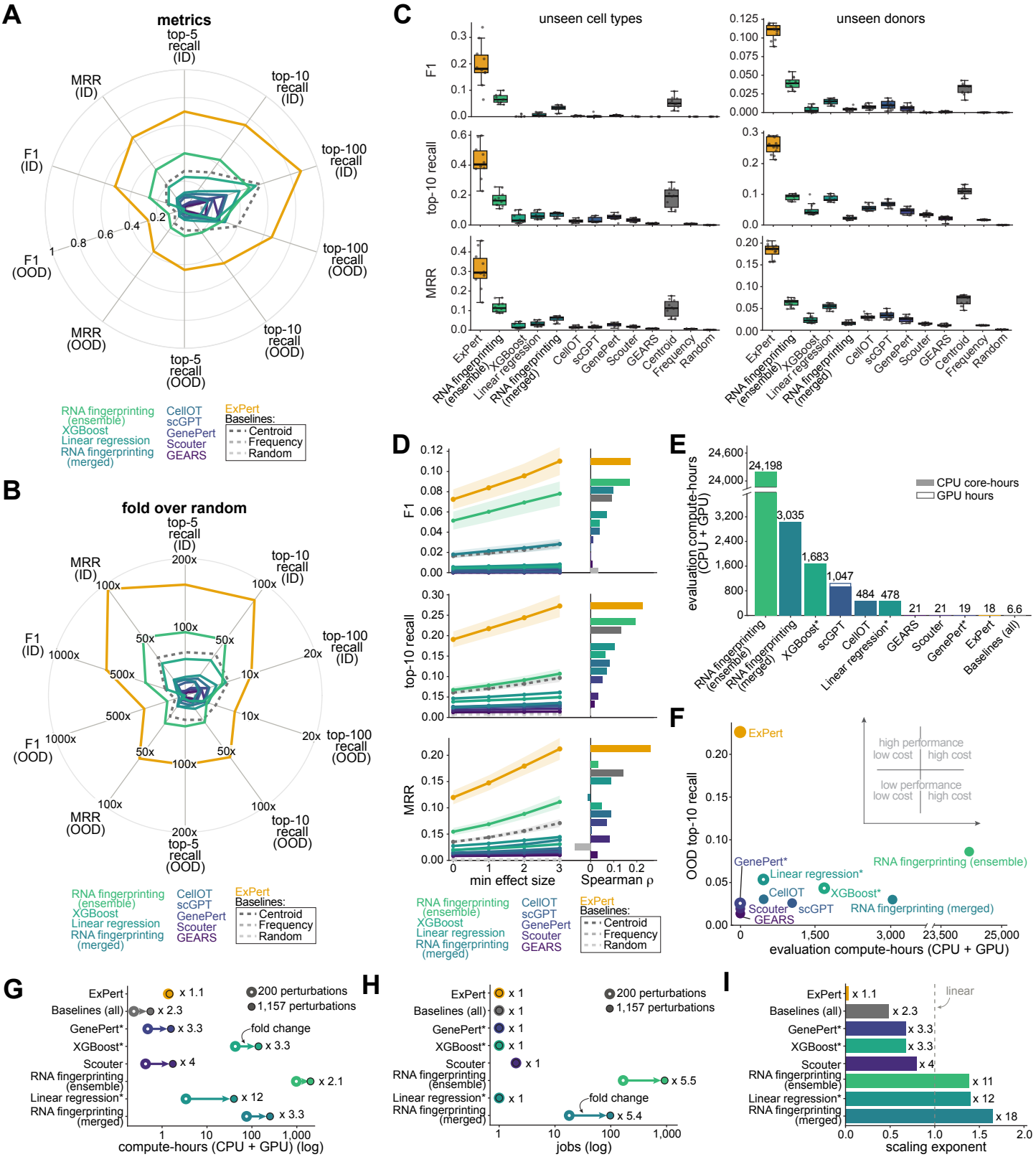
Figure 2


A
metrics
top-5 recall (ID)
top-10 recall (ID)
top-100 recall (ID)
top-100 recall (OOD)
top-10 recall (OOD)
top-5 recall (OOD)
MRR (OOD)
F1 (OOD)
MRR (ID)
F1 (ID)
RNA fingerprinting (ensemble)
XGBoost
Linear regression
RNA fingerprinting (merged)
CellOT
scGPT
GenePert
Scouter
GEARS
ExPert
Baselines:
Centroid
Frequency
Random
B
fold over random
C
unseen cell types
unseen donors
F1
top-10 recall
MRR
ExPert
RNA fingerprinting (ensemble)
XGBoost
Linear regression
RNA fingerprinting (merged)
CellOT
scGPT
GenePert
Scouter
GEARS
Centroid
Frequency
Random
D
min effect size
Spearman ρ
E
evaluation compute-hours (CPU + GPU)
CPU core-hours
GPU hours
24,198
3,035
1,683
1,047
484
478
21
21
19
18
6.6
F
OOD top-10 recall
evaluation compute-hours (CPU + GPU)
high performance
low cost
high cost
low performance
low cost
high cost
G
compute-hours (CPU + GPU) (log)
200 perturbations
1,157 perturbations
fold change
H
jobs (log)
I
scaling exponent
linear